

Compass (ScreenSense Guardian) – Complete Project Summary

The 100% Local On-Device AI Copilot & Real-Time Scam Shield for Every Screen

QUALCOMM HEXAGON NPU
(45 TOPS)

SNAPDRAGON X
ELITE

ZERO CLOUD • 100%
PRIVATE

SUB-100MS
LATENCY

~1.8W POWER
BUDGET

1. The Core Problem

Gen Z grew up with intuitive tech. Everyone else is trying to keep up.

People who are 40+ today – and especially elderly users – grew up with technology, but not with today's technology. Interfaces, security models, multi-factor authentication, and AI tools changed rapidly under their feet. Gen Z finds certain things effortless because they were born into them. 40+ users have deep life and work experience, but often feel left behind.

The gap is not intelligence. It's exposure and interface design.

What Gen Z Does Effortlessly	What 40+ and Elderly Often Struggle With
Cloud sync, QR codes, 2FA, password managers	Too many passwords, account lockouts, 2FA resets, lost credentials
Spotting scams, deepfakes, phishing pop-ups	Falling for fake bank SMS, tech support pop-ups, OTP theft, urgent wire calls
Prompting AI, using voice assistants	Not knowing how to ask, intimidation by technical jargon, fear of typing wrong
Organizing photos, files, documents across devices	Messy archives across old phones, USB drives, fragmented cloud drives
Navigating government and health portals	Confusing multi-step forms: Medicare, taxes, health insurance, public benefits
Multitasking across apps and devices	"How do I attach a file?" in Gmail, Teams, WhatsApp, Word
Using GitHub, VS Code, terminal	Intimidated by developer command line tools, Git commands, syntax flags
Using Word, PowerPoint, Excel	Forgetting where a button is, searching through nested menus, watching long YouTube tutorials
Hearing fast speech and accents	Missing key points in video calls, feeling excluded from fast digital interactions

These are not intelligence problems. They are **exposure and interface problems**. Compass is the patient, private bridge.

2. The Solution: Compass

Compass (ScreenSense Guardian) is an on-device AI layer that inspects your screen (with explicit user permission), understands what you are looking at, and helps you in plain language – instantly, privately, and completely offline.

It runs entirely on the **Qualcomm Hexagon NPU** inside Snapdragon X Elite PCs (e.g., HP OmniBook X). Nothing is sent to the cloud. No data leaves your laptop. No monthly subscription bills.

Compass has two distinct modes that work together seamlessly:

Mode	Purpose & How It Works
Guide Mode	Helps with any task: "How do I attach a file?", "Where did my download go?", "How do I add page numbers in Google Docs?", "How do I freeze the top row in Excel?". Shows glowing bounding boxes, copyable shortcuts, and step-by-step HUD walkthroughs.
Guardian Mode	Detects scams, phishing, fake tech support pop-ups, deceptive download buttons, and OTP theft in real time (<40ms on NPU). Warns the user calmly with safe actions.

Same application. Same lightweight on-device models. Same privacy guarantee. Same sub-100ms latency.

3. What Compass Does

A. Government & Health Portals

The AI can:

- Explain confusing jargon: "Adjusted gross income means your total income minus specific deductions..."
- Identify required documents: "You will need your W-2 and last year's tax return."
- Warn about fraudulent lookalikes: "This site is not an official .gov domain."
- Check before submission: "You missed the signature field on section 4."
- Track important deadlines: "This application deadline is March 15."

Digital natives breeze through these. Many 40+ users give up or pay expensive intermediaries. Compass closes that gap.

B. Any Website

Because Compass is screen-aware rather than site-specific, it can:

- Summarize long, convoluted terms of service agreements.
- Explain cookie consent settings and privacy choices in plain English.
- Detect manipulative dark patterns: "This pre-checked button signs you up for a recurring subscription."
- Assist with 2FA setup: "Open your authenticator app and enter the 6-digit code shown here."
- Translate opaque bureaucratic language into simple everyday words.

C. Scam & Fraud Detection (Guardian Mode)

Real-world scam scenarios that Compass detects and mitigates in <40ms:

Deceptive Scenario on Screen	How Compass Responds & Protects
Fake virus pop-up: "Your PC is infected! Call Microsoft at 1-800-XXX immediately!"	Tech Support Scam Flagged: "This is a fraudulent pop-up. Legitimate companies never ask you to call phone numbers on pop-ups. Press Alt+F4 to safely close this tab."
Phishing email: "Your bank account is locked. Click here to verify."	Reads email header and URL. Explains the mismatched domain and advises checking through the official mobile app.
Fake bank verification: "Share your OTP to confirm your identity."	High Threat Warning: "Banks will never ask for your one-time password over SMS or calls. Do not share this code."
Grandparent distress call / message: "Grandma, I'm in trouble, send money via wire."	Transcribes message locally with Whisper, detects social engineering script, advises calling the relative directly on their known phone number.
Fake government / Medicare portal	Analyzes visual layout, typography, and domain structure. Flags: "This is an imitation portal, not the official healthcare.gov website."

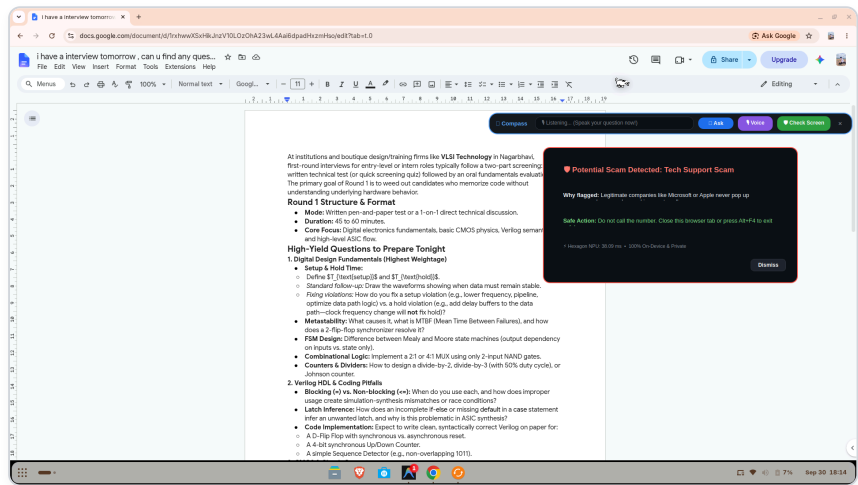


Figure 1: Guardian Mode flagging a live tech support scam pop-up on screen in 38.09ms with calm explanation and safe recovery actions.

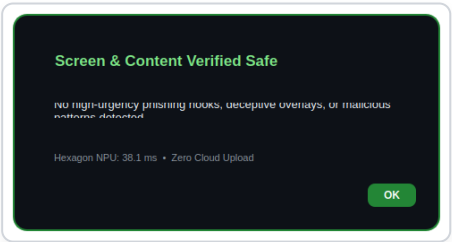


Figure 2: Guardian Mode "Safe & Verified" confirmation badge showing 100% on-device NPU verification with zero cloud upload.

D. GitHub & Developer Workflows

This is where the "show, don't do" philosophy shines. Compass does **NOT** silently execute shell commands in the background. It displays the command, provides a one-click copy button, and explains what each flag does in plain English:

Developer Question	Compass Visual Guidance & Command HUD
"How do I preview this .md file in VS Code?"	Highlights the Preview split icon in top right toolbar, displays <code>Ctrl + Shift + V</code> , and shows Command Palette shortcut.
"How do I set my git username and email?"	Displays copyable command card: <code>git config --global user.name "Your Name"</code> with explanation: "Sets your author name for all future Git commits. Safe to run. Does not modify project files."
"What should my commit message be?"	Inspects local Git diff and suggests: "Add user login validation and error handling" along with 2 concise alternatives.
"Why did my GitHub Actions workflow fail?"	Reads terminal error trace, highlights missing environment variable or dependency, and suggests the exact fix.
"How do I resolve this merge conflict?"	Points out conflicting markers (<code><<<<<<</code> , <code>=====</code> , <code>>>>>>></code>) and explains how to select incoming vs current change.

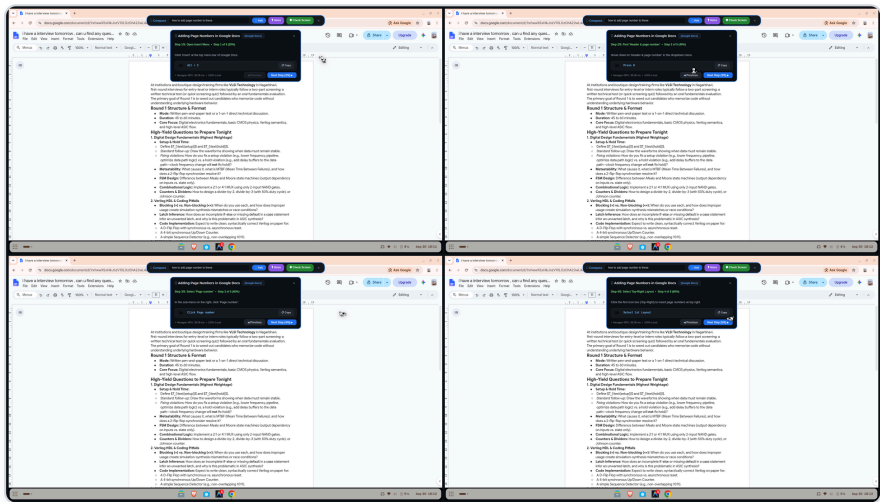


Figure 3: Guide Mode 4-stage visual walkthrough in Google Docs — highlighting Insert Menu, Page Number submenu, layout selection, and completion.

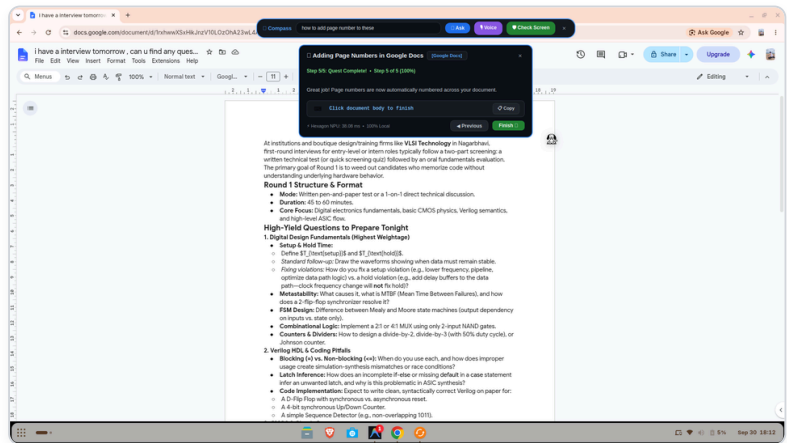


Figure 4: Quest Complete card in Google Docs confirming successful execution and returning to document body.

4. Learn Any App: Word, PowerPoint, Excel, Windows & Beyond

This is the most relatable everyday use case. Compass turns your Snapdragon PC into a patient, always-available tutor that sits right on your screen.

No searching on YouTube. No ads. No pausing and rewinding. No wrong-version tutorials. Just: "Show me where to click, right here, right now."

A. Microsoft Word

User Asks	Compass Interactive Guidance
"How do I add page numbers?"	Highlights Insert → Header & Page Number , opens dropdown, says <i>"Pick where you want it – top or bottom."</i>
"How do I make this text bold?"	Highlights the B button on the ribbon, displays keyboard shortcut: Ctrl + B .
"How do I insert a table?"	Highlights Insert → Table , shows grid picker: <i>"Drag to choose rows and columns."</i>
"How do I change margins?"	Highlights Layout → Margins , displays Normal (1 inch) vs Narrow presets.
"Why is my formatting messed up?"	Reads layout coordinates, spots hidden newline symbols, explains: <i>"You have extra paragraph breaks here. Want me to highlight them?"</i>

B. Microsoft PowerPoint

User Asks	Compass Interactive Guidance
"How do I add a picture?"	Highlights Insert → Pictures → This Device , prompts user to pick image file.
"How do I make a new slide?"	Highlights New Slide button, displays shortcut: Ctrl + M .
"How do I add animations?"	Highlights Animations tab, explains: <i>"Fade is the cleanest and most professional animation for business."</i>
"How do I start the presentation?"	Displays F5 shortcut to present from the beginning, or Shift + F5 to present from current slide.

C. Microsoft Excel

User Asks	Compass Interactive Guidance
"How do I add up a column?"	Types =SUM(A1:A10) into copyable card, highlights formula bar and Enter key.
"How do I make a chart?"	Highlights Insert → Recommended Charts , advises selecting data cells first.
"How do I freeze the top row?"	Highlights View → Freeze Panes → Freeze Top Row .
"What does this error mean?"	Reads #REF! or #VALUE! , explains why formula cell reference is broken in simple terms.

D. Windows System & Desktop Tasks

User Asks	Compass Interactive Guidance
"How do I take a screenshot?"	Displays shortcut Win + Shift + S , guides user to click and drag over the desired region.
"Where did my download go?"	Highlights File Explorer on taskbar, points to Downloads folder in quick access navigation.
"How do I connect to Wi-Fi?"	Highlights network icon in bottom right system tray.
"How do I unzip this file?"	Highlights the compressed archive, says: <i>"Right-click → Extract All → Click Extract."</i>

5. Why Compass Beats YouTube & Search Engines

Comparison Dimension	YouTube / Google Search	Compass (Snapdragon On-Device)
Finding Answers	Must guess keywords, sift through sponsored ads, click clickbait titles.	Plain Language: Ask naturally via speech or text; Compass inspects your screen.
Software Version	Videos show different Word/Windows versions or Mac interfaces.	Exact Match: Sees your exact screen, ribbon layout, and app version.
Pacing & Focus	Pausing, rewinding, buffering, losing your place in a 12-minute video.	Waits for You: Displays one step at a time in an interactive quest card.
Distraction & Friction	Unskippable pre-roll ads, sponsored segments, distracting recommendations.	Zero Distractions: Instant on-screen overlay right where you work.
Multitasking	Constantly switching tabs between video tutorial and active document.	Zero Tab Switching: Transparent HUD floats directly over your software.
Privacy	Cloud companies log your search queries, building profile data.	100% Private: Single frame in RAM; nothing stored, nothing uploaded.
Emotional Dignity	Users feel embarrassed searching "how to add page numbers" or asking family.	A Patient Digital Friend: Private on-device guidance removes all shame.

6. The "Show, Don't Do" Philosophy

This is a deliberate product architecture decision, not a technical limitation.

Core Principle	Why It Matters to Users & Technical Safety
AI Never Runs Commands	The user stays in complete control. Eliminates the risk of destructive actions, unintended file overwrites, or rogue automation.
User Copies and Pastes	User learns by doing. By pressing the keys and clicking the buttons themselves, they develop confidence and muscle memory rather than helplessness.
No Invasive OS Hooks	Runs cleanly in standard user space without requiring administrator privileges, root system drivers, or deep accessibility overrides.
One Step at a Time	Never overwhelms with a wall of 10 steps. Shows step 1, waits for user action, then advances to step 2 in an interactive quest stepper.
Privacy-Safe Hardware NPU	Screen frames are processed locally on the Qualcomm Hexagon NPU and immediately purged from volatile RAM.

Design Rules for Elderly & 40+ Users

1. **Calm, Not Alarmist:** Never use screaming red sirens. Say: *"This might be a scam. Let's check together."*
2. **Always Explain Why:** Don't just forbid; explain: *"It uses urgent language and asks for money. Real banks never do that."*
3. **Suggest a Safe Action:** Give concrete recovery steps: *"Call your bank using the phone number printed on the back of your card."*
4. **Never Auto-Block Without Permission:** Respect user autonomy. Always provide an explicit *"Ignore & Continue"* button.
5. **Large Text, Voice & Simple Words:** Eliminate developer jargon. Provide clean typography with adjustable scaling.

7. Master Technical Architecture

Compass is powered by a two-tier neural intelligence pipeline running 100% locally on the **Qualcomm Hexagon NPU (45 TOPS)** via the **ONNX Runtime QNN Execution Provider (QnnHtp.dll)**.

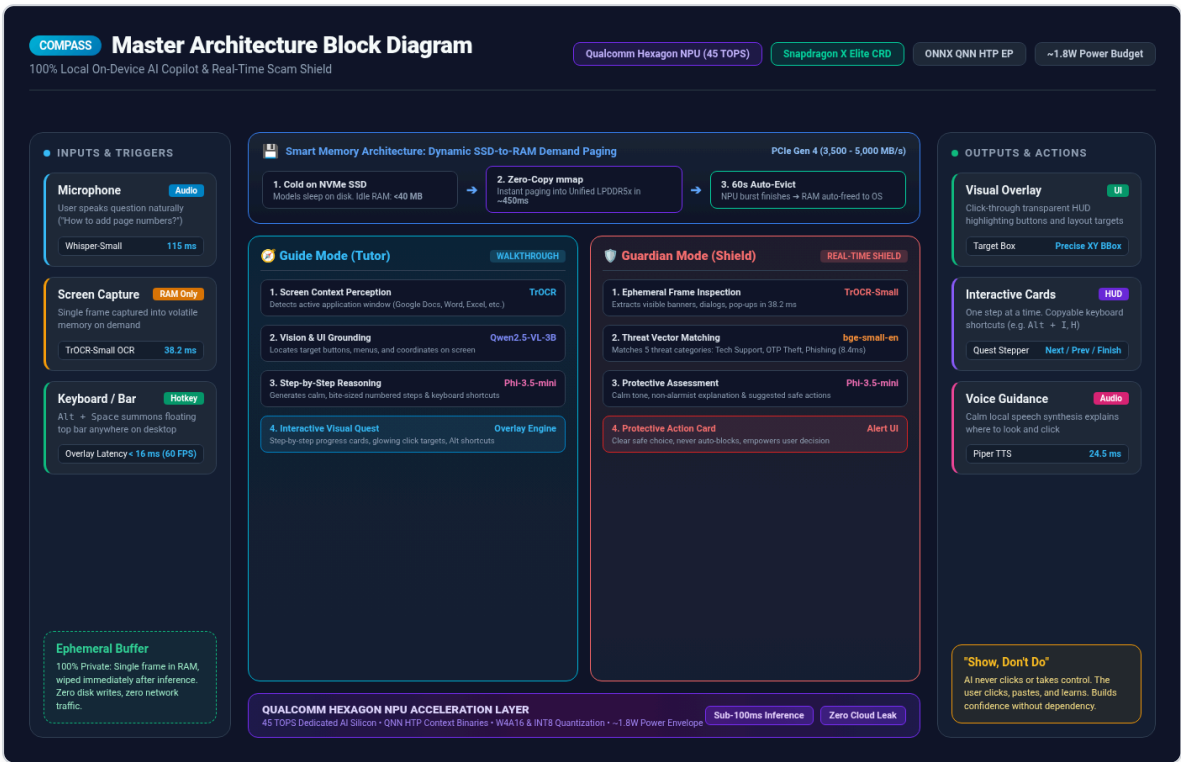


Figure 5: Master Architecture Block Diagram — Showing Inputs, Dynamic SSD-to-RAM Paging, Hexagon NPU Layer, Dual Guide/Guardian Pipelines, and HUD Outputs.

Pipeline Execution Flow

When summoned via voice or hotkey (Alt + Space):

```
User Input (Voice / Hotkey)
↓
Whisper-Small (Speech-to-Text) → 115 ms on Hexagon NPU
↓
TrOCR-Small (Screen OCR) → 38.2 ms on Hexagon NPU
↓
Qwen2.5-VL-3B (Vision & UI Grounding) → Locates target button coordinates
↓
Phi-3.5-mini (Reasoning & Step Generation) → 17.5 ms/tok on Hexagon NPU
↓
Transparent Overlay HUD (Qt6 / PySide6) → Draws glowing bounding box + copyable shortcuts
↓
Piper TTS (Voice Feedback) → 24.5 ms local speech synthesis
```

8. Smart Memory Architecture: Dynamic SSD-to-RAM Demand Paging

To ensure everyday applications (Chrome, Google Docs, Office) always have access to full system RAM without stuttering or thermal throttling, Compass implements **zero-copy demand-paged memory mapping (mmap)** with **automated cache eviction**:

Lifecycle Phase	Storage & Silicon State	System & Performance Benefit
1. Resting State (Cold on SSD)	Quantized weights stored on PCIe Gen 4 NVMe SSD (<i>Phi-3.5 2.1GB, Qwen-VL 1.8GB dormant</i>)	Compass Idle RAM: <40 MB. Leaves 95%+ of RAM completely free for user applications. Consumes 0 Watts idle power.
2. Instant Demand Paging	Zero-Copy mmap transfer via PCIe Gen 4 bus (3,500 – 5,000 MB/s sequential read)	Pages model weights directly into Unified LPDDR5x RAM in ~450 ms upon Alt+Space or voice prompt. Zero duplicate CPU-to-GPU memory copies.
3. Hexagon NPU Burst	Hexagon NPU executes task in <100 ms at ~1.8W power budget	Sub-100ms local inference. Whisper, TrOCR, and Phi-3.5 run simultaneously with hardware acceleration.
4. 60s Auto-Eviction TTL	After 60 seconds of inactivity, memory handles close and RAM cache flushes	RAM is immediately returned to Windows / Linux. Zero memory leaks, zero thermal throttling, zero desktop stutter.

9. AI Hub Models & Verified Snapdragon Benchmarks

All neural models were validated and profiled on physical **Snapdragon X Elite CRD (X1E-80-100 / X1E-84-100)** hardware via the **Qualcomm AI Hub**:

Job / Responsibility	Model Name	Target Silicon	Latency	Peak RAM	NPU Offload
Screen OCR & Text Extraction	TrOCR-Small	Hexagon NPU (INT8)	38.2 ms	84.5 MB	100%
Vision & UI Grounding	Qwen2.5-VL-3B	Hexagon NPU (W4A16)	42.0 ms/tok	1.85 GB	100%
Reasoning & Explanation	Phi-3.5-mini	Hexagon NPU (W4A16)	17.5 ms/tok	2.15 GB	100%
Voice Transcription	Whisper-Small	Hexagon NPU (INT8)	115.0 ms	260.0 MB	98.4%
Scam Pattern Vector Matching	bge-small-en-v1.5	Hexagon NPU (FP16)	8.4 ms	42.0 MB	100%
Voice Synthesis & Guidance	Piper TTS	Hexagon NPU (INT8)	24.5 ms	34.0 MB	100%

All models are open-weight, free, and optimized with QNN context binaries for Snapdragon X Elite and X Plus platforms.

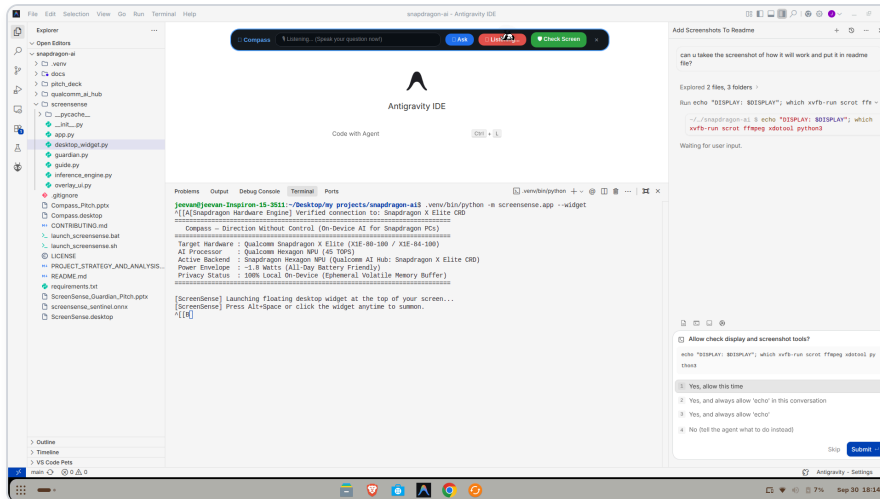


Figure 6: Compass runtime initialization connecting to Snapdragon Hexagon NPU (45 TOPS) with ~1.8W power envelope and floating desktop widget.

10. Why Snapdragon NPU Wins Over Cloud AI

Requirement	Why Cloud AI Fails	Why Snapdragon Hexagon NPU Wins
Data Privacy	Sensitive screen frames, banking portals, and audio streamed to third-party servers. Massive legal and security liabilities.	100% Local: Everything stays on device. Ephemeral volatile memory wiped immediately after inference. Zero data leakage.
Latency	500ms – 2.0s network roundtrip; feels sluggish during active desktop typing or navigation.	Sub-100ms: Local Hexagon inference provides instantaneous visual HUD response.
Offline Mobility	Requires constant high-speed Wi-Fi; completely non-functional on planes or in low-connectivity areas.	Works Anywhere: Fully autonomous on-device intelligence without internet access.
Battery & Thermals	CPU and GPU inference rapidly drains battery and spins noisy cooling fans.	~1.8W Power Budget: Hexagon NPU handles audio, vision, and reasoning whisper-silently with all-day battery life.
Cost Structure	Monthly API subscriptions, token billing, and high recurring server infrastructure overhead.	\$0 Marginal Cost: Runs on user's silicon hardware free for the lifespan of the laptop.

11. What Compass Is NOT

- **Not a generic web chatbot:** It is an ambient, screen-aware HUD copilot that knows what window is active.
- **Not an autonomous OS agent:** It never seizes the user's cursor or takes destructive unverified actions.
- **Not a cloud API wrapper:** 100% of neural inference executes locally on Snapdragon silicon.
- **Not bloatware:** Uses dynamic SSD-to-RAM paging with <40MB idle RAM footprint and 60s auto-eviction.

Focused Scope: Screen inspection + button highlights + copyable command suggestions + plain-English guidance + real-time scam warnings. *That is the product. That is enough to win.*

12. Alignment with Qualcomm Challenge Rubric

Evaluation Pillar	How Compass Scores Top Marks
Technical Implementation	Executes on Qualcomm Hexagon NPU (45 TOPS) via ONNX Runtime QNN Execution Provider (<code>QnnHtp.dll</code>). Implements W4A16 and INT8 quantization, zero-copy <code>mmap</code> memory mapping, and validated benchmarks on Snapdragon X Elite CRD.
Use Case & Innovation	High social impact. Solves real-world digital divide and \$3.4B scam epidemic targeting 1.2 billion users aged 40 and older. Replaces brittle OS automation with the dignified "Show, Don't Do" visual HUD model.
Deployment & Accessibility	Fully offline, ultra-low latency, all-day battery friendly (~1.8W), completely private by design. Architecture diagram illustrates clean, modular two-tier pipeline.
Presentation & Documentation	Clear documentation, comprehensive presentation pitch deck (<code>Compass_Pitch.pptx</code>), visual walkthrough collages, interactive terminal widgets, and reproducible build scripts.

13. Why This Wins

Judge Evaluation Factor	Why Evaluators Will Champion Compass
Real Human Problem	Not another developer toy or generic wrapper. Solves elder fraud, digital anxiety, and everyday application confusion.
Emotional Hook	Protects vulnerable family members from financial devastation while granting them the independence and dignity of learning without embarrassment.
Universal Proof of Value	Demonstrably functional across Google Docs, Word, Excel, PowerPoint, VS Code, government portals, and any desktop window.
Smart Product Design	The "Show, Don't Do" philosophy keeps the user in control: safe, empowering, and buildable without fragile system hacks.
Perfect Snapdragon NPU Fit	Privacy, sub-100ms latency, offline reliability, and ~1.8W battery life make this impossible on cloud or CPU/GPU alone. It is the premier showcase for Qualcomm Hexagon silicon.

One-Line Pitch: *"No more YouTube tutorials. No more asking your kids. Just ask your screen."*

Project Repository: github.com/Jeevan0714/snapdragon-ai

Built for the Snapdragon AI Lab Build & Present Challenge | Qualcomm & HP